

LECTURE 2b: AGENT PROPERTIES

Introduction to Multi-Agent Systems (MESIIA, MIA)

Department of Computer Science and Mathematics

Rovira i Virgili University

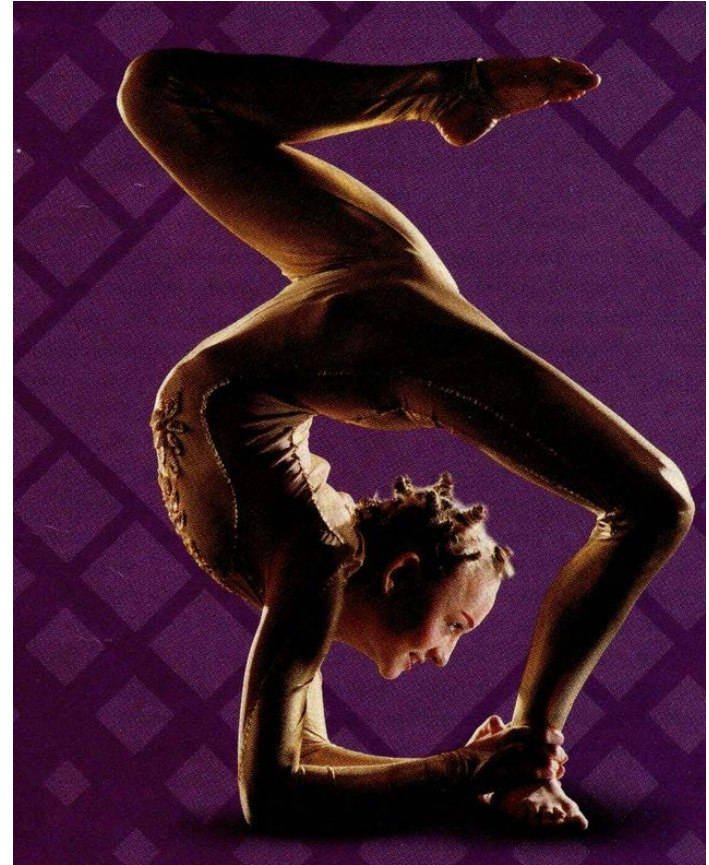
Fall 2019

Defining agents

- Many possible definitions of “agents”
 - Each author provides a set of **characteristics/properties** that are considered important to the notion of agenthood
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1-Flexibility

- An intelligent agent *is a computer system capable of **flexible** action in some dynamic complex environment*
- By **flexible**, we mean:
 - **reactive**
 - **proactive**
 - **social**



2-Reactivity

- If a program's environment is guaranteed to be fixed, the program can just execute blindly
- The real world is not like that: things change, information is incomplete. Most interesting environments are *dynamic*
 - *Multi*-agent world
- A *reactive* system is one that maintains an ongoing interaction with its environment, and responds to changes that occur in it (in time for the response to be useful)

Ways to achieve reactivity

[Recall last lecture]

- **Reactive** architectures
 - [Situation – Action] rules
 - Layered, behaviour-based architectures
 - **Deliberative** architectures
 - Symbolic world model, long-term goals
 - Reasoning, planning
 - **Hybrid** architectures
 - Reactive layer + deliberative layer [+ social layer]
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Balancing Reactive and Goal-Oriented Behaviour

- We want our agents to be **reactive**, responding to changing conditions in an appropriate (timely) fashion
- We want our agents to systematically work towards **long-term goals**
- These two considerations can be at odds with one another
- Designing an agent that can **balance** the two remains an open research problem

[recall hybrid architectures]

3-Proactiveness

- Reacting to an environment is relatively easy
(e.g., stimulus → response rules)
- But we generally want agents to *do things for us*, to act on our behalf
- Hence they must exhibit *goal-directed behaviour*
- Agents should be *proactive*



Aspects of proactiveness

- Generating and attempting to achieve **goals**
 - Behaviour not driven solely by events
 - Taking the **initiative** when appropriate
 - Executing actions/giving advice/making recommendations/making suggestions without an explicit user request
 - Recognizing opportunities on the fly
 - Available resources
 - Chances of cooperation
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Example of proactiveness (I)

- **Personal Assistant Agent**, running continuously on our mobile phone
- Location tracking
- Knows our preferences
 - Cultural activities
 - Food
- Can **proactively** warn us when we are close to an interesting cultural activity, or if it is lunch time and we are close to a restaurant that offers our favourite food [Turist@]

Crea Perfil

Fecha Inicio Viaje:
12/05/2005

Fecha Final Viaje:
12/06/2005

Interes en Arte
5

Options Exit

Crea Perfil

Interes en Historia
2

Interes en Ciencia
5

Options Exit

Personal Agent

Donde Estoy
Busca Actividades
Recomendacion
Avisos
Ver Agenda
Modifica Perfil

Select Cancel

Detalles

Nombre:
Amphitheater
Categoria:
Monumento
Tipo:
Civil
Precio:
Precio: 5
Descuentos: Pensioner
Periodo Exposicion:

Options Back

Personal Agent

Personal Agent

Nuevas Actividades
Consulta el menu
Avisos

OK

Recomendacion

Amphitheater
Sescelades
City1
Playas
Dones a Tarraco
Museu de Historia

Options Back

Foto Actividad



Options Back

turismo



Options Back

Personal Agent

Donde Estoy
Busca Actividades
Recomendacion
Avisos(!)
Ver Agenda
Modifica Perfil

Select Cancel

Avisos

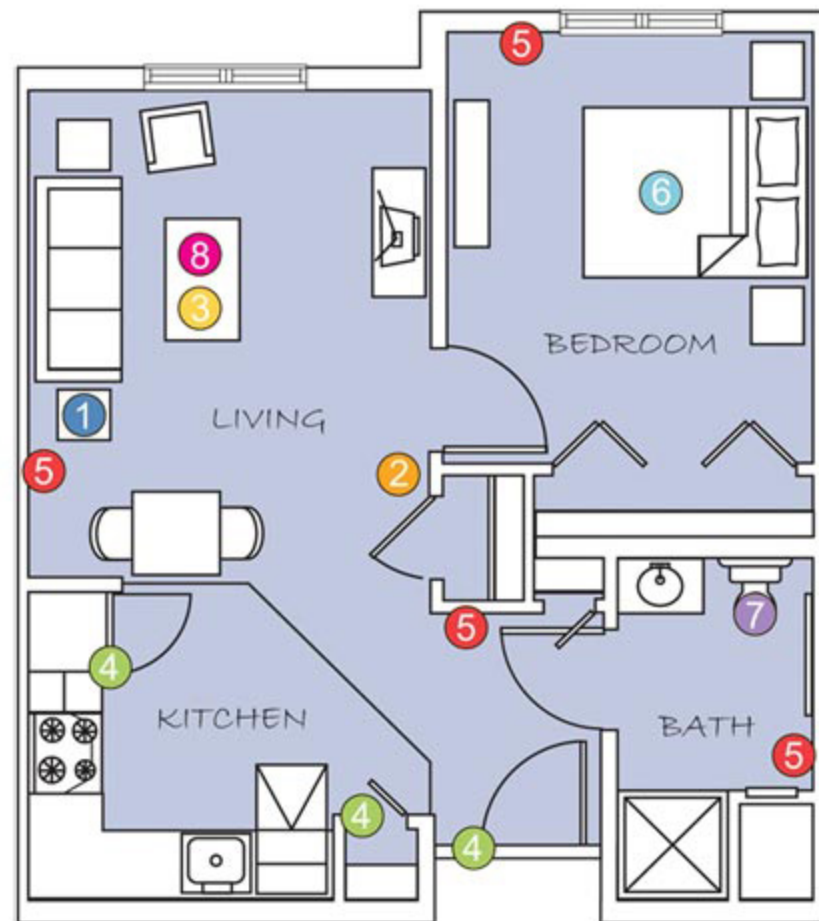
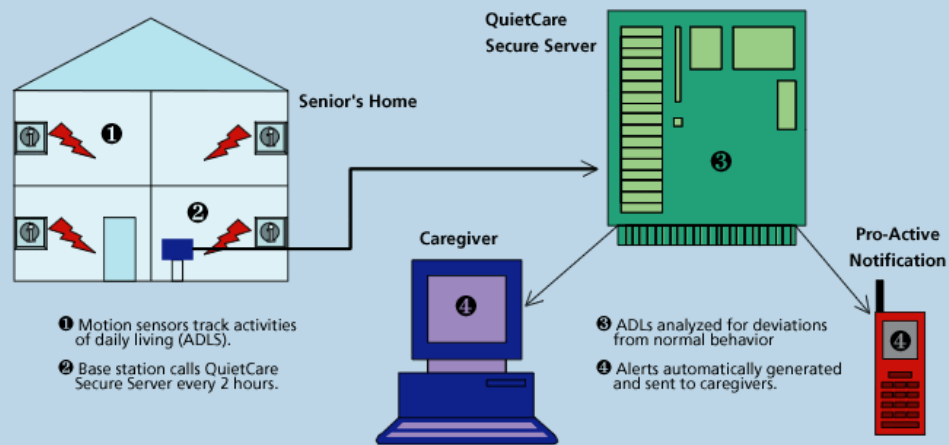
roger de flor

Options Back

Video

Example of proactiveness (II)

- Set of agents embedded in the home of an old or disabled person
- Detects the movement of the person around the house and the actions he/she performs
- Learns the usual daily patterns of behaviour
- Can detect abnormal situations, and **proactively** send warnings/alarms to family/health services
 - ❑ E.g. Too much time in the same position, long time in the bathroom, whole day without going into the kitchen, ...
 - ❑ Demo. CONFIDENCE project



- | | |
|------------------------|-----------------|
| 1 Base Station/Key Fob | 5 Motion Sensor |
| 2 Call Pendant | 6 Bed Sensor |
| 3 Home/Away Sensor | 7 Toilet Sensor |
| 4 Contact Sensor | 8 Cancel Button |

4-Social Ability

- The real world is a *multi*-agent environment: we cannot go around attempting to achieve goals without taking others into account
 - Some goals can only be achieved with the cooperation of others
 - *Social ability* in agents is the ability to interact with other agents (and possibly humans) via some kind of *agent-communication language*, and perhaps cooperate with others
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Requirements for communication

- Agent communication language
 - FIPA-ACL
 - Message types
 - Message attributes
- Agent communication protocols
- Languages to represent the content of the messages between agents
- Shared ontologies
- World-wide standards



[Lecture 4: Agent communication]

High-level activities

- Communication is a first step towards sophisticated activities:
 - Coordination
 - How to divide a task between a group of agents
 - Distributed planning
 - Cooperation
 - Share intermediate results
 - Share resources
 - Negotiation [e-commerce]
 - Agree conditions in an economic transaction
 - Find the agent that can provide a service with the best conditions (matchmaking)

[Second part of the course]

Other aspects related to communication

- Security issues

- Authentication
- Encryption

- Trust

- To what extent can we trust the other agents of the system?
 - Reputation models
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5-Rationality

- An agent will act in order to achieve its goals
- It will not act in such a way as to prevent its goals from being achieved
 - At least insofar as its beliefs permit
- For instance, it will not apply deductive procedures (blind triggering of rules) without a purpose

6-Reasoning capabilities

- Essential aspect for intelligent/rational behaviour
- **Knowledge base** with beliefs on the world
- Ability to **infer** and extrapolate based on current knowledge and experiences
- Capacity to **make plans**
- This is the characteristic that distinguishes an intelligent agent from a more “robotic” reactive-like agent

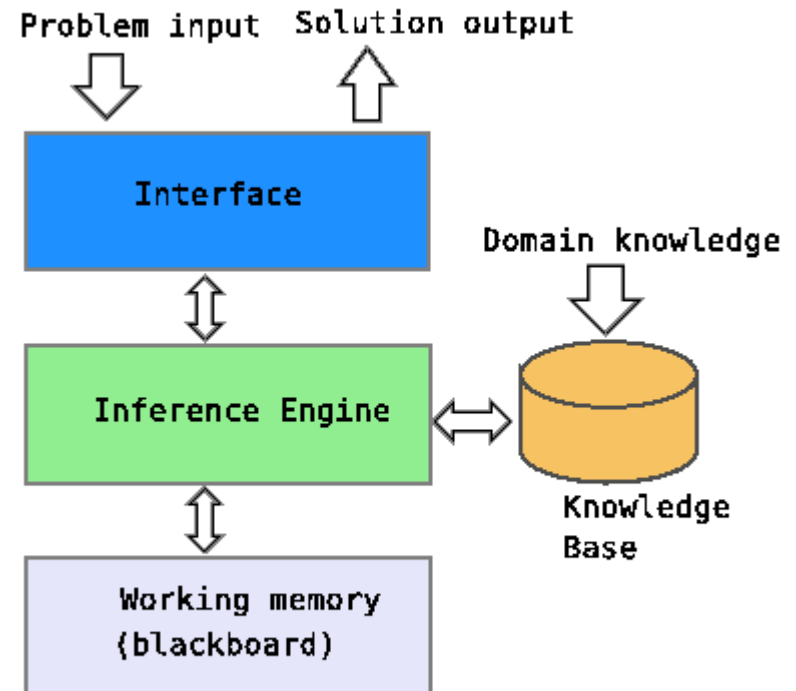
Kinds of reasoning in AI (I)

■ Knowledge-based systems / expert systems

- Reasoning techniques specialised in the system's domain
- Forward-chaining, backward-chaining, hybrid

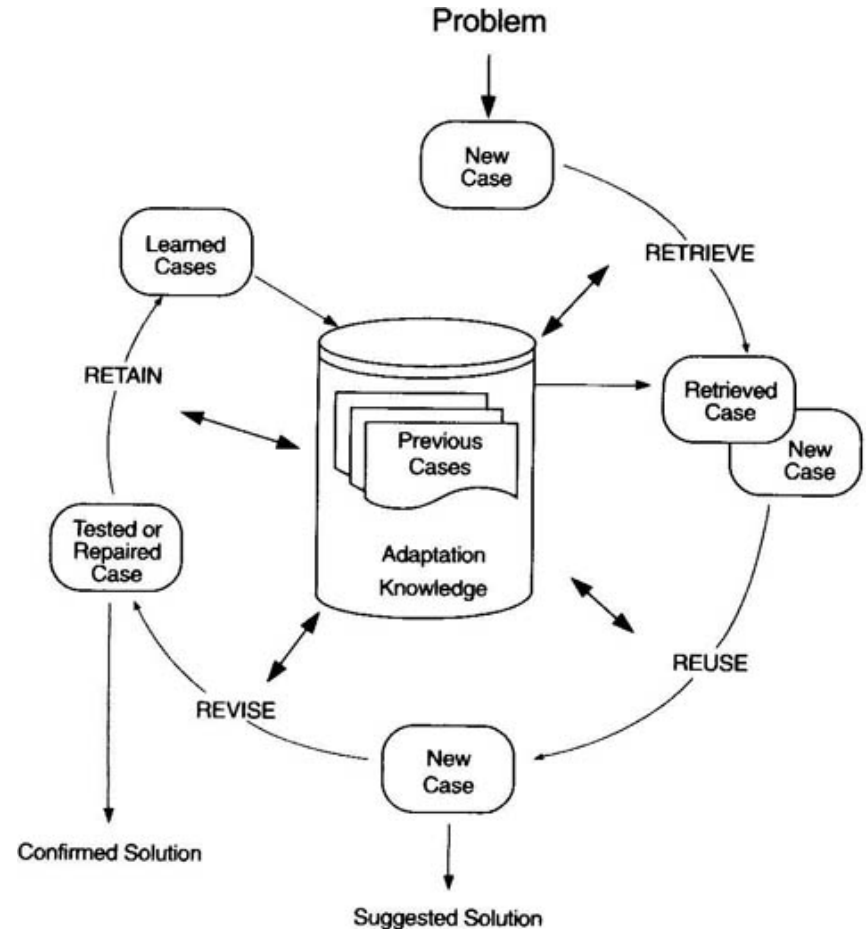
■ Rule-based systems

- Knowledge is represented as a set of rules
- Detect – Select – Apply execution cycle



Kinds of reasoning in AI (II)

- Case-based reasoning
 - Using similarity to solved problems
- Approximate reasoning
 - Fuzzy logic, Bayesian networks, probabilities, etc [PAR]



7-Learning

- Basic component for an intelligent behaviour
 - It can be considered as the [automatic] **improvement** of the performance of the agent over time
 - **Machine Learning**: large area within AI
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Ways to improve

- Make less mistakes
 - Do not repeat computations performed in the past
 - Find solutions more quickly
 - Find better solutions
 - Solve a wider range of problems
 - Learn user preferences and adapt the behaviour accordingly
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Learning tourist profiles in Turist@

- The tourist may fill an initial questionnaire
 - Analyze the tourist queries
 - E.g. Science-fiction films
 - Analyze tourist votes
 - Museum of Modern Art: very good
 - Cluster tourists with similar preferences
 - Recommend activities highly valued by tourists in the same class
-

Advantages of learning systems

- Can adapt better to a dynamic environment, or to unknown situations
 - Without the need of an exhaustive set of rules defined at design time
 - Can leverage previous positive/negative experiences to act more intelligently in the future
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8-Autonomy

- Very important difference between agents and traditional programs
 - Ability to pursue goals in an **autonomous** way, without direct continuous interaction/commands from the user
 - Given a vague/imprecise goal, the agent must determine the best way to attain it
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Autonomous decisions

- Given a certain goal ...
 - Which actions should I perform?
 - How should I perform these actions?
 - Should I seek/request/(buy !) help/collaboration from other agents?
 - Less work for the human user !!!
-

Autonomy requirements

- To have autonomy, it is necessary for an agent ...
 - To have a **control on its own actions**
 - An agent cannot be obliged to do anything
 - To have a **control on its internal state**
 - The agent's state cannot be externally modified by another agent
 - To have the appropriate access to the **resources** and capabilities needed to perform its tasks
 - E.g. access to Internet, communication channels with other agents

Autonomy limitations

- Sometimes the user may **restrict** the autonomy of the agent
 - For instance, the agent could have autonomy to search in Internet for the best place to buy a given book ...
 - ... but the agent could not have the autonomy to actually *buy* the book, using the credit card details of the user

Example: HAL

Issues

- Autonomy also raises complex issues:
 - Legal issues
 - Who is the responsible of the agent's actions?
 - Ethical issues
 - To what extent should decisions be delegated to computational agents?
 - E.g. agents in medical decision support systems
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9-Temporal continuity

- Agents are **continuously running** processes
 - Running active in the foreground or sleeping/passive in the background until a certain message arrives
- Not once-only computations or scripts that map a single input to a single output and then terminate

10-Mobility

- Mobile agents can be executing in a given computer and, at some point in time, **move physically through a network** (e.g. Internet) to another computer, and continue their execution there
- In most applications the idea is to go somewhere to perform a given task and then come back to the initial host with the obtained results

Example of use: access to a DB

- Imagine there is a DB in Australia with thousands of images, and we need to select some images with specific properties
 - We have to make some computations on the images to decide whether to select them or not
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Option 1: remote requests

- The agent in our computer makes hundreds of requests to the agent managing the DB in Australia
 - ❑ Continuous connection required
 - ❑ Heavy use of the bandwidth
 - ❑ All computations made on our computer
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Option 2: local access

- Establish connection
 - Send a specialised agent to the Australian computer holding the DB
<connection can be dismissed here>
 - The agent makes local accesses to the DB, analysing the images there
<re-establish connection>
 - Our agent comes back with the selected images
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Problems of mobile agents

■ Security

- How can I accept mobile agents in my computer? (virus !!!)

■ Privacy

- Is it secure to send an agent with the details of my credit card, or with my personal preferences?

■ Technical management

- Each computer has to be able to “pack” an agent, send it to another machine, receive agents from other machines, “validate” them, and let them execute locally.
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Kinds of mobility

- Do not confuse

- Mobile agents

- Agents that can move from one computer to another

- Agents running in mobile devices

- Agents executing in portable devices such as PDAs, Tablet PCs, portable computers or mobile phones

11- Other properties ...

■ Benevolence

- An agent will always try to do what is asked of it

■ Veracity

- An agent will not knowingly communicate false information

■ Character

- Agents must seem honest, trustable, ...

■ Emotion

- Agents must exhibit emotional states, such as happiness, sadness, frustration, ...

Relationships between properties

- More learning $\Rightarrow$ more reactivity
 - More learning $\Rightarrow$ more autonomy
 - More reasoning $\Rightarrow$ more rationality
 - More reasoning $\Rightarrow$ more proactivity
 - Less autonomy $\Rightarrow$ less proactivity
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Conclusions

- It is almost impossible for an agent to have all those properties !!!
- Most basic properties:
 - Autonomy
 - Reactiveness
 - Reasoning and learning
 - Communication

Task in activity 1 of practical exercise: think about the properties you want your agents to have !!!

Extra material for this week

- M.Wooldridge: *An introduction to MultiAgent Systems* – *beginning chapter 2, section 10.3*
- Paper on *Turist@*
- [Video] *Agent properties* – M.Wooldridge